Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Expand $(x-3)^2$.
2. Factorise x^2-16.
3. Solve $(x-2)(x+5)=0$.
4. Solve $x^2=9$ in $\mathbb{R}$.
5. Give the sign of $2x-6$.
6. Find the minimum of $(x-1)^2+4$.

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Present two methods for factorising a quadratic. In the proof using the discriminant, name the three cases and explain why they are exhaustive.

Investigate before formalising

A rectangle has perimeter 24 units. Choose several positive widths that give a rectangle. For widths 2, 4, 6, 8 and 10, calculate the length and area. Which width seems to give the greatest area? State a conjecture, then seek a justification valid for every admissible width.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.